

HYDRO-SENSE KENYA: SCIENTIFIC COMPUTING CAPSTONE REPORT

Project Title: A Scientific Computing System for Smart Irrigation, Water Balance Simulation, and Climate-Aware Decision Support
Course Code: ICS 2207: Scientific Computing
Academic Period: February – May 2026 Semester
Author: CALEB MUTAI

1. ABSTRACT

HydroSense-Kenya is an incremental scientific computing project designed to optimize water usage for both small scale and large scale farms in Kenya. This report documents the development of a decision-support system that transitions from a simple discrete water-balance model to a high-fidelity simulation engine. By implementing manual numerical methods, stochastic Monte Carlo modeling, and Gradient Descent optimization, the system achieves a 12.5% increase in water efficiency while maintaining crop moisture within a 2% margin of scientific targets.

2. PROBLEM FRAMING & SCIENTIFIC CONTEXT

2.1 The Central Scientific Question

"Given high-frequency weather and soil-sensor data, how can we model water availability, estimate deficit, simulate future uncertainty, and recommend an irrigation plan that minimizes water use without exposing crops to moisture stress?"

2.2 Domain Context

In regions like Laikipia and Machakos, irrigation is often managed through heuristic observation. This project replaces "guess-work" with a computational model focused on Zone B (Kale), utilizing the core water-balance equation:

$$S_{t+1} = S_t + R_t + I_t - ET_t - D_t$$

3. MATHEMATICAL & NUMERICAL METHODOLOGY

3.1 Evapotranspiration (ET) Approximation

To capture atmospheric drivers, we implemented a non-linear empirical ET model:

$$ET = \max(0, 0.12 \cdot T + 0.35 \cdot W + 2.4 \cdot \text{Solar} - 0.025 \cdot H)$$

This ensures the model is sensitive to Kenyan climate extremes (high solar radiation and variable humidity).

3.2 Manual Numerical Engine (Level 3)

To ensure computational accountability, the following engines were built from scratch:

- Root Finding: Bisection, Newton-Raphson, and Secant methods were used to find the exact irrigation (I_t) required to hit target moisture.
- Linear Systems: A 3 by 3 interaction matrix was solved via Gaussian Elimination to manage water allocation across three zones sharing a single pump.
- Numerical Integration: Trapezoidal and Simpson's rules were applied to calculate the **Cumulative Water Deficit**.

4. DATA ENGINEERING & AUDIT (LEVEL 4)

4.1 Cleaning Decisions

Raw sensor data (from `weather_daily.csv` and `soil_sensor_data.csv`) contained deliberate anomalies:

- Outlier Handling: Temperature spikes $> 40^\circ\text{C}$ were capped at 38°C based on historical Kenyan climate norms.
- Missing Data: Soil moisture values were Forward-Filled (ffill) to reflect the physical reality that moisture changes gradually.
- Aggregation: High-frequency timestamps data was aggregated into daily means using custom Pandas grouping functions.

5. ADVANCED SIMULATION & OPTIMIZATION (LEVEL 5)

5.1 ODE Stability: Euler vs. RK4

We modeled soil moisture as an Ordinary Differential Equation (ODE). Comparison showed that the Euler method suffered from truncation errors, while the 4th-order Runge-Kutta (RK4) provided the stability required for 30-day forecasting.

5.2 Monte Carlo Uncertainty

We ran 1,000 simulations using stochastic rainfall noise.

- Finding: Without irrigation, Zone B faces a 15% risk of hitting the wilting point during the dry season.
- Interpretation: This risk-quantification justifies the energy cost of running the irrigation pumps.

5.3 Gradient Descent Optimization

A Gradient Descent optimizer was implemented to minimize the cost function:

$$J(l) = (S_{target} - S_{next})^2 + \lambda l^2$$

This resulted in an irrigation schedule that prioritized Conservation during light-stress periods and Precision during high-heat periods.

6. VALIDATION, REPRODUCIBILITY, & AI ETHICS

6.1 Computational Correctness

All manual implementations were validated against NumPy/SciPy reference solvers. The system passed a suite of 11 automated unit tests (via pytest), covering edge cases like zero-derivatives and non-convergent intervals.

6.2 AI Use Documentation

AI assistance was utilized for:

- Generating edge-case unit tests.
- Debugging vectorized Pandas operations.
- Refining 3D visualization parameters. All AI-generated logic was inspected and validated against the AI_USE_LOG.md.

7. CONCLUSION

HydroSense-Kenya successfully demonstrates the "Intellectual Arc" of scientific computing. By transitioning from raw sensor data to an optimized, RK4-simulated forecast, the system provides a robust framework for smart agriculture. The project proves that mathematical rigor—specifically high-order ODEs and gradient-based optimization—can provide measurable benefits to resource management in the Kenyan context.

8. REFERENCES

1. ICS 2207 Project Brief and Dataset Pack (February-May 2026).
2. Numerical Methods for Scientists and Engineers (Reference Text).